

COORDINATED WORK SCHEDULES AND THE GENDER WAGE GAP*

German Cubas, Chinhui Juhn and Pedro Silos

This paper studies how coordinated work schedules across jobs contribute to the gender wage gap. Using US time diary data, we construct occupation-level measures of coordinated schedules. Higher coordination is associated with higher wages and a larger gender wage gap. Empirically, women with children allocate more time to household care and are penalised for missing work during peak hours. An equilibrium occupational choice model generates a gender wage gap of 8.9%; most of the gender wage gap is within occupations. If coordination is set to the value of healthcare support across all occupations, the within-occupation gender gap halves.

Balancing work and family is a challenge in modern societies. Household production not only limits the total number of hours that can be devoted to market work, it may also conflict with *when* hours can be supplied. For example, as long as parents cannot perfectly substitute child care responsibilities across different hours of the day, there will be temporal restrictions on when parents can supply labour. From the perspective of the employer, *when* work happens may be important too if there is joint production and firms need to coordinate workers. The nature of production may require workers to be at work at the same time to perform a joint task, raising the productivity of hours supplied when others are present. To the extent that women have more household care responsibilities than men, and therefore have greater difficulty committing to be present at any particular hour, the need for coordination impacts women more than men. This paper presents a general equilibrium model where the need for coordinating schedules generates gender wage gaps. This model is taken to the data to uncover how much of the gender gaps across and within occupations are explained by this coordination friction.

Using the American Time Use Survey (ATUS), we document novel facts regarding the timing of work for men and women. We find that parents perform household care (child care plus adult care) throughout the day—even during peak hours—suggesting that parents are indeed unable to postpone household production to non-work times. A comparison between men and women, even among full-time workers, shows that women provide more household care and work less throughout the day relative to their male counterparts. The gap in hours is small, but our point is that even small gaps can generate productivity losses, depending on the timing.

* Corresponding author: German Cubas, University of Houston, 3623 Cullen Boulevard, Room 204 Houston, TX 77204-5019, USA. Email: germancubas@gmail.com

This paper was received on 3 July 2020 and accepted on 24 November 2022. The Editor was Kjell Salvanes.

The data and codes for this paper are available on the Journal repository. They were checked for their ability to reproduce the results presented in the paper. The replication package for this paper is available at the following address: <https://doi.org/10.5281/zenodo.7336171>.

This paper has greatly benefited from the comments and suggestions of Kjell G. Salvanes (editor) and three anonymous referees. We also thank Julio Garin, Gueorgui Kambourov, Henrik Kleven, Matthias Doepke, Michele Tertilt and Ryan Michaels, as well as seminar participants at CAFRAL-Reserv Bank of India, Cal State-Fullerton, Drexel, Fudan University, Disparities in the Labor Market-FED Board, Iecon-FCEA, LACEA-LAMES, Midwest Macro, NASMES, SED, SOLE, Universidad Católica de Uruguay, University of Georgia, Universidad de Montevideo, Families and the Macroeconomy-Mannheim, Rutgers, LSE, U.S. Census Bureau, Seoul National University, University of Oklahoma, CMSG (Montreal) 2019, INSPER and FGV-EESP. Saumya Rana provided excellent research assistance.

We also employ the ATUS to measure coordinated work schedules at the occupation level. For each occupation, we record at what time of the day individuals report being at work. We associate more bunching (work schedules concentrated at particular times) with stronger coordination needs, because bunching implies that individuals are at work at the same time. We find that the degree of bunching of work hours varies across occupations and, consistent with the notion of coordination, our measure is positively correlated with other occupational characteristics such as ‘face to face discussions’ and ‘establishing and maintaining interpersonal relationships’ reported in the O*NET database.

We then use individual level data from the Current Population Survey (CPS) to study the relationship between wages and our occupational level measure of coordination. We find that our measure of coordination commands a wage premium: a one SD higher ratio leads to approximately 11% higher wages. In addition, it generates a gender wage gap: women who work in coordinated occupations are paid a higher wage, but relatively less than men (by about 5%). Interestingly, we find that married men with full-time working spouses (who presumably have greater household care responsibilities) also experience a wage penalty in high coordination occupations relative to men with non-working spouses.

Motivated by these facts, we develop a theory of occupational choice and time allocation during the day to household care and market work. In the model a gender wage gap is generated by the interaction of three key elements. First, women assign a higher value to household care—an assumption that we justify as a reflection of current social norms. Since household care activities performed at different times are less than perfect substitutes, women end up allocating more time to household care when everyone else is working, which is costly. This penalty rises with the occupational coordination needs, so women are less likely to select into occupations with higher coordination needs.

We parameterise the model with the data used in the empirical analysis. We restrict the sample to married men and women with children who are full-time workers. The model generates a gender wage gap of 8.9% (approximately 40% of the observed gender gap). We decompose the gender wage gap into the between- and within-occupation components. As in the data, most of the gender wage gap is within occupations. The within component predicted by the model is 7.2%, which accounts for 30% of that observed in the data.

To understand the extent to which occupational differences in coordination are responsible for the observed gender gap, we conduct a counterfactual exercise where coordination needs are equal across all occupations and set to the level of ‘healthcare support’—an occupation with a relatively low level of coordination. In this case, the overall gender wage gap in the model falls to 6.4%. The gender wage gap within occupations decreases by 72% to 2.0%. In another counterfactual, we reduce the difference in the value that men and women place on household care. We can think of this experiment as a way to evaluate changes in social norms that drive a reduction in the gender gap in household care responsibilities. As a result, the gender wage gap within occupations decreases by 50% to 3.6%. Finally, we focus our study on the effects of the ability to substitute household care during the day. Our baseline calibration points to an economy in which household care activities are fairly substitutable, but imperfectly so. They may reflect parenting styles or just constraints on the time of the day in which some activities take place (for example, meetings with school teachers). We thus analyse a counterfactual economy in which women can now more easily distribute the household care to off-peak times so they do not incur a productivity loss. As a result, the gender wage gap within occupations decreases by 31% to 5%.

A large literature in macroeconomics and labour economics relates family arrangements and the labour supply of its family members. Important contributions are Albanesi and Olivetti (2009), Doepke and Tertilt (2016) and Bick and Fuchs-Schündeln (2018). Our paper is also closely connected to the literature that examines the role of frictions on workers' labour supply responses. These frictions could arise from fixed wage-hour packages offered by employers that result in non-linear payment schedules. Important contributions are Prescott *et al.* (2009), Rogerson and Wallenius (2009), as well as Rosen (1976), Altonji and Paxson (1988), Altonji and Paxson (1992) and Blundell *et al.* (2008). Recent papers have emphasised the role of coordination as the driving force behind non-convex budget sets. The wage-hour combinations available to workers may be sparse due to the needs for coordination. This need may arise at the firm level or even at a more aggregate, economy-wide level. For example, Guner *et al.* (2014) studied how the Spanish work schedule with long lunch breaks affects parental time allocation. Moreover, Guner *et al.* (2019) found that the inflexibility of work schedules partially explains the low fertility observed in some rich countries. Other recent papers study the labour supply responses to changes in, for instance, taxes or other economic conditions. They find stark differences between responses with non-linear payment schedules and those predicted by linear payment schedules and an absence of coordination. Examples include Chetty *et al.* (2011), Rogerson (2011) and Labanca and Pozzoli (2022). We contribute to this literature by exploring how coordination requirements influence labour supply, as well as another important margin—occupational choice. Instead of examining labour supply responses to tax changes, we examine how these hour requirements driven by coordination needs conflict with the demands of household production and, consequently, lead to the gender wage gap.¹ Our paper measures the coordination needs indirectly, and thus it does not use direct evidence on the coordination technology in production. However, recent work provides evidence on the coordination and transmission of complex information between co-workers in specific production processes and working environments. Examples are Battiston *et al.* (2020), Gibbs *et al.* (2021) and Labanca and Pozzoli (2022).

Our work is also closely related to the literature that relates occupation-specific characteristics to the gender wage gap. Goldin (2014) argued that much of the remaining gender wage gap can be explained by the lack of flexible work arrangements. Along these lines, a number of papers have shown that the gender gap is particularly large in jobs that demand long hours (Gicheva, 2013; Cha and Weeden, 2014; Cortes and Pan, 2016a,b; Duchini and Effenterre, 2017; Erosa *et al.*, 2017 and Wasserman, 2019).² Compared to these studies, our focus is on flexible timing, rather than the flexibility to set the number of hours. We show that, while the demand for long hours and our measure of coordination are positively correlated, the correlation is far from perfect and both contribute to the gender wage gap.³

A recent paper, Mas and Pallais (2017), elicit workers' willingness to pay for flexible schedules using a field experiment. These authors found that, while the average willingness to pay for

¹ Our occupational choice model integrates the timing of work with the timing of household care in a unified framework where family responsibilities play a key role. This feature differentiates our work from previous works that study the timing and synchronisation of work schedules. Examples of these studies are Weiss (1996), Hamermesh (1999), Cardoso *et al.* (2012) and Eden (2017).

² The requirement for long hours has also been associated with less flexible work schedules. Thus, our work also relates to Goldin and Katz (2011), Flabbi and Moro (2012) and Wiswall and Zafar (2017).

³ A recent paper by Denning *et al.* (2019) showed that the positive relationship between hours worked and earnings is virtually absent within occupations and it is only observed across occupations. This finding suggests that the hours gap between men and women cannot account for the within-occupation gender wage gap if this hours penalty is applied. In our paper we show that differences in the within-occupation gender wage gap can be large even when the gender hours gap is small.

flexibility is low, there is also a long right tail in the willing to pay distribution suggesting compensating differential for inflexibility still could be large at the margin.⁴ Another recent paper, Chen *et al.* (2019), estimates the value of flexibility among drivers of the ride-sharing platform Uber. Drivers have almost total flexibility when to supply labour, to the point of being able to react on an hourly basis to unexpected shocks to their reservation wage. The authors estimate the surplus from that flexibility to be large, and hence their results are roughly in line with our findings.

The paper is organised as follows. Section 1 describes our data, the temporal patterns of work and household care in the ATUS, as well as our measure of hours bunching that proxies for coordination requirements. Section 2 presents the model. Section 3 illustrates the model mechanics with simple examples. Section 4 reports our reduced-form regression results using individual-level CPS data. Section 5 describes the calibration and our counterfactual experiments. Section 6 concludes.

1. Time Allocation by Gender, and Coordinated Work Schedules

1.1. Data

We base our analysis on the 2003–14 ATUS-CPS data (Bureau of Labor Statistics, 2014 and Flood *et al.*, 2022). One respondent per household is drawn from the Current Population Survey samples, and the interviews are conducted two to five months after the last CPS interview. The ATUS respondent is asked to fill out a time diary over the previous day, recording their activities and starting and ending times. There are 17 aggregate activities and we focus on two activities, ‘work and work-related activities’ and ‘caring for and helping household members’.⁵ For each individual, we calculate minutes spent on these activities for each hour of the day using information on starting and ending times. The ATUS also contains demographic and labour force information, including labour force status and usual hours worked. We restrict our sample to adults who are 18 to 65 years old. Our main sample of time diary respondents consists of 106,620 observations. For comparing time use of men and women, we focus on full-time workers (those whose usual weekly hours are greater than or equal to 35).⁶ The full-time worker sample consists of 66,023 observations. We do not make restrictions based on self-employment status and also include multiple job holders. However, to construct the ratio of hours worked in the 8 to 5 time interval (which we label *ratio8to5*) at the occupation level, we include only full-time workers who worked a minimum of 35 hours in their main job. For the regression analysis where we explore the impact of occupation-level *ratio8to5* on wages, we include all individuals in the

⁴ They also found that workers particularly dislike working evening and weekend shifts, which at first appears to be counter to our story. Occupations that require evening and weekend shifts (such as security guards) may appear to be flexible in terms of our bunching measure, but this may just be a reflection of a 24-hour production cycle. To address this issue, we re-run our regressions, controlling for the share of workers who are shift workers in the occupation. We find that our results are robust to adding these additional controls. We show this result in Section A.1.5 of the Online Appendix.

⁵ We do not include the aggregate category, ‘household activities’, which includes housework. If we include ‘household activities’, the gender gap is considerably larger. In most of our analysis we take a conservative route by restricting our attention to the aggregate category ‘caring for and helping household members’ that only includes child care and elder care reported as the primary activity. In Table 3 we also explore a more expanded definition of child care such as ‘socialising and communicating’ when a child is present.

⁶ The fact that women are more likely to work part time and part-time workers are paid a lower hourly wage is well established. We focus on full-time workers in our study to push home our point that schedules and when work happens matter for hourly wages in addition to the number of hours worked.

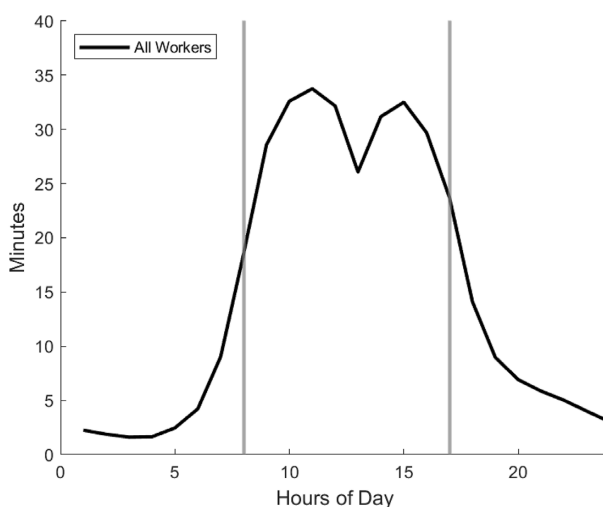


Fig. 1. *Work among Full-Time Workers.*

Notes: Data are from the 2003–14 ATUSs. The figure is based on 8–65-year-old ATUS respondents who report being full-time workers in the activity summary file. The figure reports average minutes spent by hour of the day on ‘work and work-related activities’ on the diary day.

CPS, including those who are not time-use survey respondents. For this case, the sample sizes are considerably larger, with the sample consisting of 263,313 individuals who are full-time workers aged 18 to 65 with non-missing weekly wages. Since the time use surveys are conducted three months after the main CPS interviews, we use variables such as age and work status that are collected at the time of the time use survey whenever possible. Some of the information, however, such as education, is available only in the main CPS data. Online Appendix A.1 contains more detail regarding construction of our data.

1.2. *Timing of Work and Household Care*

In this section we describe patterns of time use over the course of a single day for full-time workers by gender, marital status and parental status. These patterns show how time allocated to market work is constrained by the demands of family time and how those constraints differ for men and women. Figure 1 explores *when* work happens. The figure graphs the average number of minutes worked by one-hour time bins for full-time workers. The figure shows that most (74%) work occurs during the 8 a.m. to 5 p.m. interval with a break between 12 p.m. to 1 p.m.⁷ Figure 2 graphs the average number of minutes worked by marital and parental statuses. The left panel shows work for married individuals, men and women, with at least one own child in the household, who work full-time. The right panel shows work for singles with no children. Even among full-time workers, women work less than men, with the gap being largest among those married with children. Table 1 further explores the gender differences in work for this group. The table shows that women work approximately 0.9 hours less on weekdays and 0.7 hours less on

⁷ Average minutes worked per hour is well below 60, which may reflect the fact that we are averaging over all seven days of the week, including weekends. We do not intend to eliminate weekends, as work also takes place on those days. In addition, work over weekends varies across occupations.

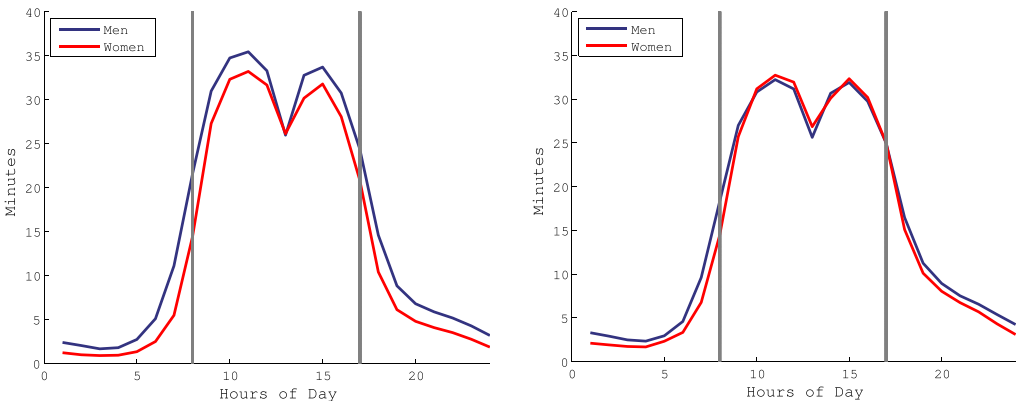


Fig. 2. Work among Full-Time Workers.

Notes: Data are from the 2003–14 ATUSs. The figure is based on 18–65-year-old ATUS respondents who report being full-time workers in the activity summary file. The figure reports average minutes spent by hour of the day on ‘work and work-related activities’ on the diary day. Both weekdays and weekends are included. The left panel includes workers who are married with a spouse present with at least one own child in the household. The right panel includes workers who are single and without children.

Table 1. Work among Full-Time Workers, Married with Children.

	Weekday	Weekend	Weekday			
Female gap in work hours	−0.898*** (0.0694)	−0.749*** (0.0674)	−0.901*** (0.0692)	−0.911*** (0.0702)	−0.703*** (0.0698)	−0.490*** (0.0768)
Observations	12,113	12,344	12,113	12,113	12,113	8,393
Day of week and year			x	x	x	x
Education, age, race and # Kids				x	x	x
Usual weekly hours					x	x
Usual weekly hours less than 50						x
Average hours, men	7.904	2.163				
Average hours, women	7.006	1.414				
Average hours, total	7.611	1.906				

Notes: Data are from the 2003–14 ATUSs. The table is based on 18–65-year-old ATUS respondents who report working full time in the activity summary file. We keep those who are married with a spouse present and have at least one own child in the household. The dependent variable is total hours spent on ‘work and work-related activities’ on the diary day. Each column reports the coefficient on the ‘female’ dummy. Column (5) controls for usual weekly hours worked reported in the activity summary file. Column (6) only includes workers who reported usual weekly hours of less than 50. Individual observations are weighted by ATUS weights for multi-year data files. *** $P < 0.01$.

weekends. Column (5) controls for usual weekly hours worked reported in the activity summary file. Column (6) only includes workers who reported usual weekly hours less than 50. Both of these restrictions reduce the gap in hours worked, but even among full-time workers who work less than 50 hours, married women with children work almost 0.5 hours less on weekdays relative to their male counterparts.

Figure 3 graphs the temporal pattern of household care among full-time workers who are married with children (left panel) and singles without children (right panel). The differences in the temporal pattern of work and household care, however, are notable. Both women and men with children report household care with noticeable bumps up in the early morning and evening hours. The temporal pattern of care for full-time workers with children is negatively related to the temporal pattern of work, with the fewest minutes devoted to care activities during the 8 to 5

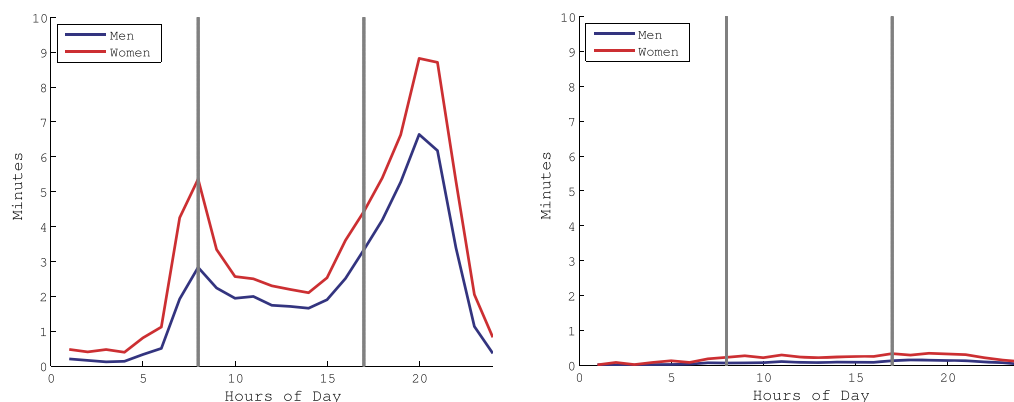


Fig. 3. *Household Care among Full-Time Workers.*

Notes: Data are from the 2003–14 ATUSs. The figure is based on 18–65-year-old ATUS respondents who report being full-time workers in the activity summary file. The figure reports average minutes spent by hour of the day on ‘caring for and helping household members’ on the diary day. Both weekdays and weekends are included. The left panel includes workers who are married with a spouse present with at least one own child in the household. The right panel includes workers who are single and without children.

Table 2. *Household Care among Full-Time Workers, Married with Children.*

	Weekday	Weekend	Weekday			
Female gap in household hours	0.436*** (0.0276)	0.264*** (0.0332)	0.436*** (0.0276)	0.348*** (0.0270)	0.319*** (0.0272)	0.266*** (0.0327)
Observations	12,113	12,344	12,113	12,113	12,113	8,393
Day of week and year			x	x	x	x
Education, age, race and # kids				x	x	x
Usual weekly hours					x	x
Usual weekly hours less than 50						x
Average hours, men	0.821	1.002				
Average hours, women	1.257	1.267				
Average hours, total	0.963	1.093				

Notes: Data are from the 2003–14 ATUSs. The table is based on 18–65-year-old ATUS respondents who report working full time in the activity summary file. We keep those who are married with a spouse present and have at least one own child in the household. The dependent variable is total hours spent on ‘caring for and helping household members’ on the diary day. Each column reports the coefficient on the ‘female’ dummy. Column (5) controls for usual weekly hours worked reported in the activity summary file. Column (6) only includes workers who reported usual weekly hours of less than 50. Individual observations are weighted by ATUS weights for multi-year data files. *** $P < 0.01$.

interval. However, even during the 8 to 5 interval, household care does not fall to zero. Table 2 shows that among married men and women with children, women engage in nearly 0.5 hours more household care during weekdays and 0.3 hours more on weekends. Different controls reduce the gap, but the table shows that women significantly allocate more time to household care than men.

Table 3 provides further detail regarding differences in hours of child care provided by mothers and fathers. First we examine detailed care categories adopting a method introduced by Stewart (2010). Stewart (2010) defined three broad categories of child care: ‘routine’, ‘enriching care’ and ‘other’. Included in routine care is physical care and looking after children. ‘Enriching care’ includes activities such as reading to children and playing sports with children. ‘Other’ includes more nebulous activities such as ‘organising and planning for household chil-

Table 3. *Household Care Activities of Parents by Marital and Work Statuses (Hours).*

Activity	Females			Males
	Married NW	Married FT	Single FT	Married FT
<i>Panel A: weekday</i>				
With children aged less than six				
Routine	1.4 (0.03)	1.0 (0.03)	0.8 (0.04)	0.4 (0.02)
Enrichment	0.6 (0.02)	0.4 (0.02)	0.3 (0.02)	0.3 (0.01)
Other	0.4 (0.01)	0.4 (0.01)	0.4 (0.02)	0.1 (0.01)
With children aged 6–18 only				
Routine	0.4 (0.01)	0.2 (0.01)	0.2 (0.01)	0.1 (0.01)
Enrichment	0.6 (0.02)	0.4 (0.01)	0.3 (0.02)	0.3 (0.01)
Other	0.4 (0.01)	0.2 (0.01)	0.3 (0.01)	0.1 (0.01)
<i>Panel B: weekend</i>				
With children aged less than six				
Routine	1.0 (0.02)	0.9 (0.02)	0.7 (0.03)	0.4 (0.02)
Enrichment	0.4 (0.02)	0.4 (0.02)	0.3 (0.02)	0.3 (0.01)
Other	0.1 (0.01)	0.1 (0.01)	0.1 (0.01)	0.1 (0.00)
With children aged 6–18 only				
Routine	0.2 (0.01)	0.1 (0.01)	0.1 (0.01)	0.1 (0.00)
Enrichment	0.5 (0.02)	0.4 (0.01)	0.4 (0.02)	0.3 (0.01)
Other	0.1 (0.01)	0.1 (0.01)	0.1 (0.01)	0.1 (0.00)

Notes: Data are from the 2003–14 ATUSs. The table shows the average hours allocated to household care activities by parents. See the text for the definitions of the main activities. See the Online Appendix for detailed activities that are included in each category. ‘Married NW’ refers to married women with a spouse present who are not working, ‘Married FT’ refers to men and women who are married with a spouse present and working full time, ‘Single FT’ refers to single women who are working full time.

dren’, ‘attending children’s events’ and ‘picking up and dropping off children’. Table A.4 in the Online Appendix provides the full list of activities included in each of the three broad categories.

Table 3 compares the hours of each type of care performed by non-working married mothers, full-time married mothers, full-time single mothers and full-time married fathers, respectively. The top panel reports hours during weekdays while the bottom panel reports hours during weekends. The table also separates out households where at least one child is under the age of six (school age). Looking at hours of routine care in families with young children, we see (not surprisingly) that non-working mothers provide the most care, 1.4 hours, while full-time married mothers and full-time single mothers provide 1.0 and 0.8 hours, respectively. Full-time married fathers provide considerably less, 0.4 hours. What is surprising is that non-working and full-time working mothers provide the same amount of care in the ‘other’ category—all three groups provide 0.4 hours on a typical weekday. There are differences when we examine households

with only older children, but our main point is that the child care provided by non-working and full-time working mothers is not as different as one might have thought, especially when it comes to the non-routine care categories. This type of child care does not constitute a lot of hours, but the table shows that certain activities cannot be easily outsourced.

To summarise, we showed that married women with children who are full-time workers report fewer hours of work in the time diary data relative to their male counterparts—a phenomenon we call ‘missing hours’. The ‘missing hours’ occur throughout the day and is distinct from women being less likely to work long hours. Correspondingly, married women with children also perform more household care than men. The extra household care is unlikely to be routine child care since we are considering full-time working women. Instead, the extra hours are likely to consist of a catch-all ‘other’ category that includes such activities as organising and planning, driving children, attending doctors’ appointments and children’s activities. These activities add up to a small number of hours, but are likely to entail costly work interruptions.

1.3. Measure of Coordinated Work Schedules

Building on the previous section, we construct our measure of coordinated work schedules for different occupations. Call the time intervals between 12 a.m. and 8 a.m., between 8 a.m. and 5 p.m. and between 5 p.m. and 12 a.m. A , B and C , respectively. Then A_{ij} , B_{ij} and C_{ij} refer to the sum of minutes worked by individual i in occupation j in those respective intervals. We sum over individuals to get occupation-level equivalents

$$A_j = \sum_{i=1}^{M_j^A} w_i A_{ij}, \quad B_j = \sum_{i=1}^{M_j^B} w_i B_{ij}, \quad C_j = \sum_{i=1}^{M_j^C} w_i C_{ij},$$

where w_i refers to the survey weight of the individual and M_j^A , M_j^B and M_j^C are the numbers of individuals in occupation j in intervals A , B and C , respectively.

Our measure of coordinated work schedules at the occupation level is the ratio of minutes worked in the 8 to 5 interval relative to total minutes worked:

$$ratio_{8to5_j} = \frac{B_j}{A_j + B_j + C_j}.$$

We include only full-time workers in calculating this ratio. A higher ratio indicates that a greater amount of work in the occupation occurs during the standard 8 to 5 work day. We also standardise this measure by subtracting the mean and dividing by the SD. We view a higher ratio as indicating the need for greater coordination, with more hours worked concentrated during peak hours. We construct ratios by detailed 2002 census occupations, resulting in 493 non-missing ratios (see Online Appendix Table A.7). Online Appendix A.1 provides further details on the construction of the ratios.

We highlight some occupations in Figure 4. Among occupations with more educated workers, ‘Lawyers’ and ‘Financial analysts’ have standardised ratios of 0.588 and 0.663, respectively. ‘Writers and authors’ have a relatively low ratio of 0.157. ‘Physicians and surgeons’ have the lowest ratio of -0.010 . In occupations with relatively less educated workers ‘Nursing, psychiatric and home health aides’ has a very low ratio of -1.284 . ‘Cashiers’ have a ratio of -0.781 . On the other hand, ‘Secretaries and administrative assistants’ have a high ratio of 1.052.

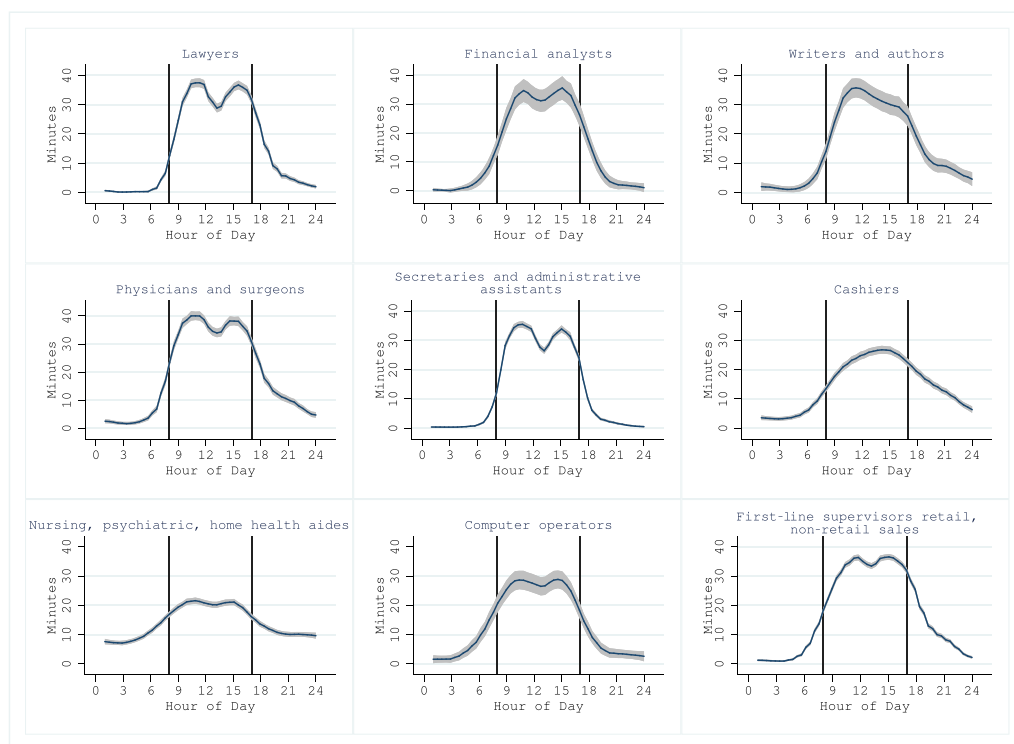


Fig. 4. *Timing of Work in Selected Occupations.*

Notes: Plots show smoothed values from local polynomial regressions of minutes spent by hour of the day on ‘work and work-related activities’ for different occupations. Shaded areas display confidence intervals. Both weekdays and weekends are included.

How is our measure related to other occupational characteristics? Table 4 reports correlations of our measure, *ratio8to5*, with other occupational characteristics reported in the O*NET database.⁸ The table shows that our measure points to the need for coordination with others in the workplace. Our measure is positively correlated with ‘establishing and maintaining interpersonal relationships’ and ‘face to face discussions’. On the other hand, it is negatively correlated with ‘assisting and caring for others’.

One can view our measure as a rather arbitrary way to think about the concentration of working hours during a day as the peak hours are fixed to be between 8 and 5. As an alternative, one could think of how concentrated the hours are during the day without pre-establishing the times of the day. We have also explored an alternative measure of concentration based on the Herfindahl index.

⁸ O*NET (O*NET OnLine, 2020) reports scores on the importance of occupational characteristics for detailed 2018 SOC. We used the census crosswalk between 2018 SOC and 2010 census occupation codes and used the 2018 American Community Survey (Ruggles *et al.*, 2022) to take weighted averages to more aggregate 2010 census occupation codes. There were also a number of changes between the 2002 and 2010 census codes. We use the number of full-time workers aged 18–65 in the ATUS as weights to aggregate O*NET measures to the 2002 census occupation codes.

Table 4. *Correlations between Importance of Occupational Characteristics and ratio8to5.*

#Cat.	Name: O*NET characteristic	Corr. coefficient
1	Assisting and caring for others	−0.1908*
2	Coaching and developing others	0.0471
3	Developing and Building Teams	0.0525
4	Establishing and Maintaining Interpersonal Relationships	0.2948*
5	Face-to-Face Discussions	0.2336*
7	Social orientation	0.0471
8	Training and Teaching Others	−0.0719
10	Guiding Directing and Motivating Subordinates	0.0501
	Concentration index	0.7430*
	Male overwork	0.1290*
#Cat.	Name: O*NET skill measures	Corr. coefficient
1	Social skills	0.2129*
2	Abstract skills	0.3632*
3	Manual skills	−0.4362*
4	Routine skills	−0.3737*

Notes: The table shows correlations between our standardised *ratio8to5* and O*NET occupational characteristics for 434 detailed census 2002 occupations. Here *ratio8to5* is the ratio of total hours worked by all full-time workers during the hours 8 a.m. to 5 p.m. relative to total hours worked in each occupation category in the ATUS time diary data. See the Online Appendix for the detailed definitions of the O*NET characteristics, as well as for details on the variables used and for matching across O*NET standard occupation codes (SOCs) and 2002 census occupation codes. The asterisk denotes significance at the 5% level.

Let $work_j^k$ be the total weighted time spent working in each day of the week-hour time bin k in occupation j ,

$$work_j^k = \sum_{i=1}^{M_j} work_{ijk} \cdot w_i,$$

where i denotes the individual in occupation j and w_i denotes the weight of individual i . Let $share_j^k$ be the fraction of the total time spent in each occupation in each time bin and each day:

$$share_j^k = \frac{work_j^k}{\sum_k work_j^k}.$$

Our concentration index measure is the Herfindahl index defined as

$$cr_j = \sum_k (share_j^k)^2.$$

Table 4 shows that our coordination measure, *ratio8to5*, and the concentration index measure are highly positively correlated, with the correlation equalling 0.743. The table also reports the correlation of our measure and the measure of ‘Male overwork’ used by Cortes and Pan (2016b). ‘Male overwork’ is defined as the fraction of male workers who report working more than 50 hours per week. This ratio is calculated for men with at least some college education. The correlation between these two measures is 0.129, indicating that, while our measure is positively related to the demand for long hours, it is by no means perfectly correlated. Thus, our measure captures another important aspect of hours requirements on the job such as the requirement to be present when others are present.

Another aspect worth exploring is the extent to which our measure of coordination is correlated with other skill requirements at the occupational level, such as social skills (Deming, 2017), and

abstract, routine and manual skills (Acemoglu and Autor, 2011).⁹ The bottom rows of Table 4 show that our measure is indeed strongly positively correlated with social skills and abstract cognitive skills, while it is negatively correlated with both manual and routine skills. In the next section, we explore the extent to which the returns to our measure of coordination is impacted when we add these skill measures as additional controls.

2. The Model

Given the evidence shown in the previous section, we present a general equilibrium model that links three ingredients. First, frictions at the occupational level imply a different productivity for labour supplied at different hours of the day. This leads to a pattern of bunching of hours that varies by occupation. Second, women value household care differently than men. And third, childcare time is not perfectly substitutable during the day. We use the model to motivate the regression specification in Section 4 below that links the gender gap to the *ratio8to5* calculated in Section 1.

2.1. Environment

The economy is populated by a continuum of male and female workers of equal masses that sum to 1. Everyone lives for one period and values consumption of a market good, denoted by c , and a home good denoted by h . People rank bundles of the two goods according to a Cobb–Douglas utility function:

$$u(c, h) = (c)^{\nu^s} (h)^{1-\nu^s}.$$

Here ν^s represents the weight of market goods in utility for gender s with $s = f, m$.

Three aspects of the preferences are worth noting. First, males and females differ in the relative value they give to the home good. This asymmetry should not be taken literally as a fundamental difference in preferences. It is a convenient way to capture observed differences in hours of household care between males and females. This difference may reflect social norms, differences in bargaining power, discrimination, etc., but an explicit modelling of these features is outside the scope of the paper. A second and related aspect is that the decision unit is the individual and not the household. The reason for doing so is data limitations. Specifically, observations in the ATUS are at the individual level, and there is no information on spousal time allocation. Third, as in our empirical part, h represents household care activities that are normally not outsourced (even for full-time workers), and thus we abstract from other household care activities including leisure. In order to isolate the effect of coordinating activities during the day, we want to focus on those activities that are directly performed by the worker.¹⁰

Workers have one unit of time, a fraction of which can be supplied in a labour market that features J occupations and that are labelled using the integer j . Occupations are mutually

⁹ For ‘Social skills’, we follow Deming (2017) and use four measures: ‘Social perceptiveness: being aware of others’ reactions and understanding why they react as they do’, ‘Coordination: adjusting action in relation to other’s actions’, ‘Negotiation: bringing others together and trying to reconcile differences’ and ‘Persuasion: persuading others to change their minds and behaviour’. We create a composite score for ‘Social skills’ by averaging the four individual scores. We also construct four other composite measures of skill requirements closely following Acemoglu and Autor (2011) and Denning *et al.* (2019). See Online Appendix A.1 for details.

¹⁰ There is evidence that women also allocate more hours than men to household care activities that we are abstracting from (e.g., preparing meals). To the extent that these activities conflict with work schedules our results provide a lower bound for the mechanism we propose in explaining the gender gap.

exclusive; workers can only work in one occupation. Workers receive a wage w_j per unit of time they supply in occupation j . Earnings from the supply of labour is how workers finance purchases of the market good c .

Time is divided into two sub-periods of equal length. We label the first period as ‘prime’ (or 1), and the second period as ‘home’ (or 2). We associate the first period in the model with the 8 a.m. to 5 p.m. period in the data.¹¹ Workers do not exclusively choose how to split their unit of time between working in the market and home care; they also choose how much to allocate to either activity during each sub-period. We respectively denote by h^i and l^i the home care and work choices in sub-period t . Since the total time used must add up to one, the following identity must hold:

$$h_{j,1}^i + l_{j,1}^i + h_{j,2}^i + l_{j,2}^i = 1.$$

Since either sub-period represents half of total time, the following must also be true:

$$h_{j,t}^i + l_{j,t}^i = 0.5$$

for each sub-period t .¹²

Our model abstracts from the labour market participation margin, as well as the possibility of working part time. We recognise that these decisions are not independent from fertility (also not modelled) and that there is a dynamic component to them. Abstracting from these margins does not mean we think they are not relevant to study the gender wage gap. Our goal is to quantify in isolation the new mechanism we propose. We find that this mechanism explains a sizeable portion of the observed gender wage gap, but not its totality.

Thus, our focus is on modelling the choice of hours during the day conditional on being a full-time worker when caring for children conflicts with the work schedule. The empirical evidence in Section 1 shows that, conditional on being full-time workers, the distribution of working hours during the day is not uniform. Workers bunch hours at particular times of the day and the degree of bunching is higher in certain occupations. This evidence, coupled with data from O*NET (see Table 4), suggests that in some occupations the need to coordinate workers’ schedules is stronger. For example, some occupations rely more on team production where workers’ tasks are complementary, while in others individuals work mostly on their own. This difference in the production technology translates into a friction on an individual’s supply of labour. If an individual’s tasks are complementary with others’ in the same occupation, not supplying labour when others do has a productivity penalty. For example, missing a team project meeting has a productivity penalty. This penalty is likely to differ across occupations. Returning to our model, while we do not explicitly model the production technologies that lead to coordination needs, we assume that not supplying labour during prime time has a penalty. More specifically, a reduced-form way of capturing the importance of coordinating workers’ schedules is given by a reduction in the effective hours of work when labour is not supplied during prime time:

$$l_j^i = l_{j,1}^i + l_{j,2}^i - (0.5 - l_{j,1}^i)^{\alpha_j} \quad \text{with} \quad \alpha_j \geq 0 \quad \text{for } j = 1, \dots, J.$$

¹¹ That our prime period starts at the beginning of the ‘day’, as opposed to the middle (as in the data), is an innocuous assumption. It is convenient and nothing of substance changes if we assume that the prime period starts in the middle of the day. Moreover, the ‘prime’ period in the model need not be equal to half of the total time endowment. The model is flexible and can be adapted to however one defines business hours to be in the data.

¹² To be clear, that we define each sub-period to be equal to 0.5 in length is a normalisation. It is akin to defining the total time endowment to be equal to 1.

The parameter α_j drives the penalty for not supplying labour during prime time in occupation j . The penalty is infinitely large when α_j is zero: any missed hours during prime time yield zero effective labour. The penalty vanishes as α_j tends to infinity, as the effective labour is the sum of $l_{j,1}^i$ and $l_{j,2}^i$, irrespective of the value of $l_{j,1}^i$.¹³

This specification allows for large productivity losses in some occupations when a worker postpones working time to the second period ('home'). The penalty is large in occupations with a low α_j , while in those with a high α_j the loss is minimal. The maximum amount of time any worker (male or female) can work in the prime period is 0.5. For a given amount of work, supplying more home care time during period 1 leads to a lower productivity per hour. The extent of the productivity loss is occupation, but not gender specific. Although α is exogenous, and thus our model is silent about the source of these differences, one interpretation is that workers coordinate because productivity rises when everyone is present. By convention, this coordination takes place during the period we call prime time.

The production of home goods employs hours both within prime and home time according to a constant elasticity of substitution (CES) technology:

$$h^i = [(h_1^i)^\rho + (h_2^i)^\rho]^{1/\rho}$$

with ρ the parameter that governs the elasticity of substitution between the supply of home care time across the two time periods. If $\rho < 1$, home care in different periods are imperfect substitutes.

On the production side, there is a set of J intermediate good producers indexed by j . We associate the production of an intermediate good with an occupation. Each produces an amount X_j of the intermediate good. Production employs a linear technology in effective units of labour N_j ; that is, $X_j = A_j N_j$, where A_j is a total factor productivity parameter.¹⁴ Markets are competitive and the producer faces prices for her good p_j and wages w_j .

The producer of intermediate good j solves the maximisation problem

$$\max_{N_j} p_j X_j - N_j w_j$$

subject to the available technology $X_j = A_j N_j$. The solution to the problem is $p_j = w_j / A_j$. Intermediate good producers sell to a final goods producer. The technology for producing a certain amount Y of the final good from a vector of quantities of intermediate services $\{X_1, \dots, X_J\}$ is described by

$$Y = \sum_{j=1}^J \{\kappa_j X_j^\beta\}^{1/\beta},$$

where β governs the elasticity of substitution between the intermediate goods and κ_j the share of each one in final production.

The final good producer solves the maximisation problem

$$\max_{\{X_1, \dots, X_J\}} \sum_{j=1}^J \{\kappa_j X_j^\beta\}^{1/\beta} - \sum_{j=1}^J p_j X_j.$$

¹³ While from a technological perspective $\alpha_j = 0$ is possible, an occupation with such an extreme need for coordination cannot exist in equilibrium. Wages would need to be infinity for workers to choose it.

¹⁴ The role of the total factor productivity parameters is only to help deliver the empirical distribution of earnings across occupations. Replicating that distribution is necessary to obtain a plausible between-occupation gender wage gap.

Note that in equilibrium $X_j = A_j N_j$ and $p_j = w_j / A_j$, so that this maximisation problem implicitly defines labour demand functions $\{N_j = N_j^d(w_j, N_{-j})\}_{j=1}^J$.

2.2. Individual's Decision Problem

Prior to choosing an occupation, each individual draws a vector of taste parameters for occupations, Ω_i , from gender-specific distributions $F(\theta_{j,s})$. Thus, each individual i is represented by the vector

$$\Omega_i = \{\theta_{i,1}, \dots, \theta_{i,J}\}.$$

Each element of the vector, $\theta_{i,j}$, represents the taste for occupation j and are independent across occupations. These shocks are important in matching the empirical sorting of workers into occupations. These shocks represent the various forces we leave out, but which nonetheless determine the size and gender composition of occupations (comparative advantages, experience, etc.).¹⁵

The amount of effective labour supplied by a worker of gender s in occupation j is l_j^s . Effective labour is compensated at a rate w_j per unit.

The value of occupation j for an individual of gender s is

$$V_j^s(\theta_j^s) = \theta_j^s \left\{ \max_{c^s, l_{j,1}^s, l_{j,2}^s, h_{j,1}^s, h_{j,2}^s} \{u(c^s, h^s)\} \right\}$$

such that

$$\begin{aligned} c^s &= l_j^s w_j, \\ h_{j,2}^s + l_{j,2}^s &= 0.5, \\ h_{j,1}^s + l_{j,1}^s + h_{j,2}^s + l_{j,2}^s &= 1, \\ l_j^s &= l_{j,1}^s + l_{j,2}^s - (0.5 - l_{j,1}^s)^{\alpha_j} \quad \text{with } \alpha_j \geq 0, \\ h_j^s &= [(h_{j,1}^s)^\rho + (h_{j,2}^s)^\rho]^{1/\rho}. \end{aligned}$$

Each individual chooses from the set of J occupations the one that yields the highest utility:

$$\hat{j}^s = \arg \max \{V_1^s, \dots, V_J^s\}. \quad (1)$$

Here V_j^s denotes a worker's value function in her chosen occupation $\hat{j}$.¹⁶ The occupational choice determines an endogenous distribution of male and female workers across occupations. Let μ_j^s denote the mass of gender s workers in occupation j ; then $\sum_{j=1}^J (\mu_j^f + \mu_j^m) = 1$. Define $\mu_j = \mu_j^f + \mu_j^m$ as the size of occupation j .

2.3. Aggregation and Equilibrium

Given wages, individuals solve the optimisation problem yielding value functions $\{V_j^s\}_{j=1}^J$.

¹⁵ The majority of elementary school teachers are women—despite that occupation's high *ratio8to5*—but very few work as police officers (despite a low *ratio8to5*). See Table A.7 in the Online Appendix for the fraction of females and *ratio8to5* for the occupations used in the empirical analysis. It is outside the scope of this paper to investigate all the reasons, beyond the coordination of schedules, why women sort into or out of some occupations. The taste parameter shocks are a convenient way of capturing these reasons.

¹⁶ To save on notation, we write V_j^s for the value of working in occupation j . It is understood that this value depends on a taste parameter θ_j .

For an occupation j , its population satisfies $\mu_j^s = \text{Prob}(V_j^s > V_{-j}^s)$, where we define the vector V_{-j}^s to be equal to $\{V_1^s, \dots, V_{j-1}^s, V_{j+1}^s, \dots, V_J^s\}$.

For occupation j , the total labour input is defined as

$$N_j = \frac{\mu_j^m}{\mu_j} (l_{j,1}^m + l_{j,2}^m - (0.5 - l_{j,1}^m)^{\alpha_j}) + \frac{\mu_j^f}{\mu_j} (l_{j,1}^f + l_{j,2}^f - (0.5 - l_{j,1}^f)^{\alpha_j}).$$

In addition, in equilibrium

$$w_j = \kappa_j A_j^\beta N_j^{\beta-1} \sum_{j=1}^J [\kappa_j X_j^\beta]^{1/\beta-1}.$$

Given an individual's occupational and hour choices, our model predicts the *ratio8to5* for working hours for each occupation, which we denote as *ratio8to5_j*. Following the definition of these indicators presented above, its model counterpart is given by

$$\text{ratio8to5}_j = \frac{\mu_j^m}{\mu_j} \frac{l_{j,1}^m}{(l_{j,1}^m + l_{j,2}^m)} + \frac{\mu_j^f}{\mu_j} \frac{l_{j,1}^f}{(l_{j,1}^f + l_{j,2}^f)},$$

where μ_j is the fraction of workers in occupation j , and μ_j^m and μ_j^f are the fractions of males and females in occupation j , respectively.

Before we take this model to the data and examine the role of coordination frictions in accounting for the gender gap, we illustrate the model's mechanisms using a simpler version than that described above.

3. Model Mechanics in a Simple Case

We restrict attention to an economy with only two occupations. We provide a numerical example choosing illustrative values of the parameters to uncover the main mechanisms. We analyse three environments that differ in the degree of heterogeneity among workers, detailed below. There is a set of parameters that are common across these economies. We assume a Cobb–Douglas technology for the final good, and earnings in each occupation represent an equal share in final aggregate income, i.e., $\kappa_1 = \kappa_2 = 0.5$. The parameters that govern the productivity penalty due to the coordination of workers are $\alpha_1 = 0.8$ and $\alpha_2 = 2.8$. In other words, in occupation 1 coordination is much more important. Table 5 summarises the results of each of the experiments that are described below.

3.1. Economy 1: Homogeneous Agents without Gender Differences

This economy features a mass of size 1 of workers who have the same weight for market consumption: $\nu_m = \nu_f = 0.8$. The parameter driving the elasticity of substitution between home care time at the two time periods of the day (between h_1 and h_2), ρ , is set to 0.6. The results are shown in panel A of Table 5.

Consumption goods and household care are substitutes. More market consumption implies more market time and less time allocated to household care. The equilibrium features sorting into occupations, with a larger mass of workers choosing occupation 2. Because of the higher α , productivity losses due to coordination are smaller in occupation 2. As a result,

Table 5. *A Simple Case with Gender Differences.*

Occupation (1)	% Workers (2)	Bunching ratio (3)	Earnings (4)	$l_1 + l_2$ (5)	l (6)	% Females (7)	E. Gap (8)
<i>Panel A: no gender differences</i>							
1	0.48	0.59	0.42	0.82	0.80		
2	0.52	0.51	0.38	0.80	0.80		
<i>Panel B: gender-specific u</i>							
1	0.44	0.55	0.46	0.91	0.90	0	
2	0.56	0.52	0.36	0.73	0.73	89	1.005
Gender earnings gap				1.031			
<i>Panel C: gender-specific ν and tastes</i>							
1	0.50	0.60	0.40	0.81	0.80	50	1.047
2	0.50	0.51	0.40	0.81	0.80	50	1.005
Gender earnings gap				1.026			

occupation 2 is more attractive. Despite the higher cost, the final goods technology rules out an equilibrium in which no one chooses occupation 1. Wages adjust to leave workers indifferent between the two occupations. The higher wage results in a higher supply of labour in occupation 1. Hence, $l_1 + l_2$ is larger. However, they have to pay a higher penalty and, as a result, effective hours are equal across occupations. To summarise, workers in occupation 1 supply more market work and less household care. The opposite is true in occupation 2. Why is $ratio_{8to5}$ higher in occupation 1? This is because workers, in an attempt to minimise the hours penalty, bunch hours to a larger extent in the prime period. Prime time cannot be exclusively devoted to work, however, because home care cannot be substituted perfectly across the two sub-periods. Finally, since workers in occupation 1 devote relatively more hours to work in prime time, they end up devoting relatively more hours of household care during home time (h_2).

3.2. *Economy 2: Gender Differences in Household Care Responsibilities*

We now consider the case of an economy where males and females are differentiated by the weight in market consumption ν . Half of the workers have $\nu = 0.9$ (male) and half have $\nu = 0.7$ (female), i.e., females have stronger preferences for household care. The results are shown in panel B of Table 5.

Because of their different preferences, females and males do not sort randomly into the two occupations. Females have a relatively higher preference for household care and thus they populate only occupation 2, the high- α occupation. Occupation 2 allows females to supply household care without paying too high an hours' penalty. In addition, since household care hours are complementary during the day, more total household care time means a higher supply of household care hours both within prime and home time, i.e., household care hours need to be smoothed during the day. Occupation 2 allows them to do that at a relatively lower cost. Males have a comparative advantage in occupation 1. Because they want to supply more labour, they downplay the importance of the penalty when choosing their occupation. As a result, a higher proportion of males work in occupation 1.

To summarise, workers in occupation 2 spend a bit more time in home care (because the wage is lower). Consequently, raw hours, effective hours worked and earnings are all lower. Therefore, in equilibrium there is a gender gap in earnings per hour of 3%.¹⁷ Females in occupation 2 represent 89% of the workers in that occupation. Because their time allocation is different, males and females earn different amounts per hour with a wage gap of 0.5%. Thus, in this example most of the gender wage gap is due to earnings per hour differentials between occupations (as opposed to within occupations).

3.3. *Economy 3: Gender Differences in Household Care Responsibilities and Tastes for Occupations*

We now consider the case of economy 2, but we incorporate gender differences in tastes for each occupation that results in 50% of workers being female in each occupation. The results are shown in panel C of Table 5.¹⁸

Conditional on working in occupation 1, females want to work more than if they are in occupation 2 since they want to minimise the coordination cost. However, they work less than males since they want to supply relatively more time to household care. As a result, they end up paying a higher cost in terms of effective hours, and thus their earnings per hour are going to be lower than males. This is also the case for males and females in occupation 2, but the effects are lower given that α is higher. For this reason, the gender earnings gap per hour is higher in occupation 1. In equilibrium, this example features a gender gap in earnings per hour of 5% in occupation 1 and no gender earnings gap in occupation 2. The aggregate gender earnings gap for this economy is also 3%. While the aggregate earnings gap is the same as in economy 2, the gender gap in this economy is entirely driven by earnings differences within occupations due to the fact that women have fewer effective hours.¹⁹

As in the other economies, conditional on being in occupation 1, workers want to supply more time in prime time to minimise the coordination cost, and as in the other cases, the ratio is higher in occupation 1. Therefore, the example reflects the negative correlation between *ratio8to5* and the gender earnings gap we find in the empirical part of the paper.

4. Coordinated Work Schedules and the Gender Wage Gap

The model analysed in the previous section links earnings to *ratio8to5* through worker sorting and general equilibrium effects. Since frictions at the occupational level affect genders differently, we are interested in examining how earnings by gender interact with *ratio8to5* (a proxy for frictions at the occupation level). The stylised parametric assumptions we adopted in the previous sections suggest that the model is capable of generating the following evidence we observe in the data: (a) occupations with higher *ratio8to5* pay more and (b) the gender gap is larger in occupations with a higher *ratio8to5*. In this section we analyse how our measure of coordinated work schedules (*ratio8to5*) is priced in the labour market, and how it impacts the gender wage

¹⁷ Throughout the paper, the terms gender gap in earnings per hour and gender wage gap are used interchangeably.

¹⁸ In the Online Appendix, we report similar experiments with a model of two-earner households. In the paper we work with single-agent models because we lack data on time allocation for both spouses.

¹⁹ Note that in this particular example the share of workers in each occupation is 50%, so all the differences in earnings per hour come from differences in effective hours and not from differences in the wage rates across occupations.

Table 6. *Gender Gap in Log Weekly Earnings by Coordination Measure ratio8to5.*

	(1) Baseline	(2) (1) + agg. education	(3) (2) + overwork	(4) (3) + O*NET
<i>Panel A: all</i>				
Female	−0.218*** (0.0194)	−0.254*** (0.0129)	−0.244*** (0.0129)	−0.212*** (0.0118)
<i>ratio8to5</i>	0.113*** (0.0170)	0.0644*** (0.0136)	0.0631*** (0.0138)	0.0692*** (0.0132)
Female × <i>ratio8to5</i>	−0.0490* (0.0264)	−0.0457** (0.0180)	−0.0436** (0.0190)	−0.0279* (0.0155)
Observations	2,63,245	2,63,245	2,63,179	2,56,672
<i>Panel B: single without children</i>				
Female	−0.135*** (0.0157)	−0.169*** (0.0117)	−0.165*** (0.0118)	−0.132*** (0.0113)
<i>ratio8to5</i>	0.102*** (0.0146)	0.0623*** (0.0132)	0.0613*** (0.0131)	0.0620*** (0.0105)
Female × <i>ratio8to5</i>	−0.0216 (0.0207)	−0.0254 (0.0162)	−0.0242 (0.0168)	−0.0099 (0.0142)
Observations	73,536	73,536	73,516	71,577
<i>Panel C: married with children</i>				
Female	−0.263*** (0.0229)	−0.298*** (0.0159)	−0.286*** (0.0162)	−0.255*** (0.0137)
<i>ratio8to5</i>	0.109*** (0.0189)	0.0652*** (0.0142)	0.0649*** (0.0151)	0.0722*** (0.0150)
Female × <i>ratio8to5</i>	−0.0626* (0.0327)	−0.0595** (0.0221)	−0.0582** (0.0236)	−0.0400** (0.0192)
Observations	1,10,230	1,10,230	1,10,206	1,07,626

Notes: Standard errors are clustered at the occupation level and given in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$. See the text for an explanation of the control variables.

gap. Specifically, we estimate the following regression at the individual level:

$$\ln W_i = \beta_0 + \beta_1 \times female_i + \beta_2 ratio8to5_j + \beta_3 female_i \times ratio8to5_j + \beta_4 X_i + \varepsilon_i.$$

Here $\ln W_i$ is the log of individual weekly earnings, $female_i$ is the female dummy, $ratio8to5_j$ is the ratio of hours worked in the 8 to 5 interval that varies at the occupation level j , the X_i are other observable characteristics including dummies for race and education and a quartic function in age. We also control for (log) hours worked last week so that the coefficients we report reflect gaps in the hourly wage.²⁰ Our sample includes only full-time workers. Variable β_1 measures the impact of the female dummy, β_2 measures the impact of working in occupations with a more concentrated work day and β_3 captures how being female interacts with working in these occupations.

Table 6 reports the results of the regression. The top panel reports the results for all full-time workers. Column (1) presents the baseline results. Women earn on average 22% less than men. Individuals in occupations with higher *ratio8to5* earn higher wages, with a one SD higher ratio leading to approximately 11% higher wages. The interaction term indicates that women suffer about a 5% higher penalty in these occupations. In column (2) we control for occupation-level

²⁰ Specification with constructed log hourly wages where we divide weekly earnings by usual weekly hours yielded very similar results.

education that reduces the size of the wage premium associated with these occupations and also the female-specific penalty. In column (3) we control for the fraction of male workers in the occupation who report working more than 50 hours per week—the measure of ‘overwork’ used by Cortes and Pan (2016b). In column (4) we include O*NET skill measures such as social, abstract, manual and routine skills. The coefficient on the concentration measure is still significant, although the addition of skill measures reduces the female-specific penalty somewhat.

The bottom two panels report results separately by marital and parental statuses. Panel B reports results for single men and women. Notably, the interaction terms are all insignificant, pointing to the fact that there is no penalty for women associated with coordinated work schedules. Panel C reports results for married men and women with children. The female interaction terms are larger and significant, which suggests that the results pooling over all workers reported in the top panel were largely due to the married with children group. Notably, even when we include ‘overwork’ and O*NET skill measures as additional controls, the coefficient on the interaction term is -4% and still significant at the 5% level.

These regressions indicate that workers in occupations where most adhere to a standard 8 to 5 schedule are paid a higher wage. However, the gender gap in these occupations is larger. This pattern is particularly pronounced when we restrict our sample to married men and women with children, strongly suggesting that conflicts related to work and family time play an important role.

One objection to our interpretation of the results is that employers may be practising statistical discrimination against married women with children and the level of discrimination is particularly severe in occupations with coordinated schedules. This alternative interpretation, while closely related, suggests that it is not necessarily the temporal constraints that women face due to household care responsibilities that are at play. To further investigate this alternative explanation, we examine different groups of married men (with children) who are full-time workers, differentiated by the work status of their wives.²¹

In Table 7 we investigate whether these constraints imposed by care responsibilities translate into wage penalties. Table 7 reports the results of a regression in a similar format as Table 6, but we now make comparisons among men only. The sample includes all married men with children matched to a spouse in the CPS data. The variables ‘Spouse PT’ and ‘Spouse FT’ are indicators equal to 1 if the wife works part time or the wife works full time, respectively. The omitted category is ‘Wife not working’. The coefficients indicate that married men with full-time working wives earn approximately 5% less than married men with non-working wives, indicating either selection or specialisation effects.²²

The coefficients of interest however are the interaction terms that indicate that a one SD higher ratio leads to a 3%–4% higher penalty for men with part-time working wives, and a 5%–6% penalty for men with full-time working wives. Table 7 shows that the phenomenon is not unique to comparisons between men and women, but is more general and applies where there is balancing between work and household care. It is possible that the direction of causality is the opposite—that spouses with less earning power engage in more household care. Regardless, our

²¹ Online Appendix Table A.2 and Table A.3 examine work hours and household care of married fathers by work status of the spouse. The tables show that fathers with full-time working spouses work 0.25 fewer hours on a weekday and perform somewhere between 0.07 to 0.12 more hours of household care on a typical weekday relative to fathers with non-working spouses.

²² In the regressions shown in Table 7 we control for log weekly hours on the main job to focus on the hourly wage gap. Regression results using constructed hourly wages (weekly earnings divided by usual weekly hours) yield very similar results.

Table 7. *Log Weekly Earnings of Males by Working Status of Spouse and Coordination Measure ratio8to5—Married with Children.*

	(1) Baseline	(2) (1) + agg. education	(3) (2) + overwork	(4) (3) + O*NET
<i>ratio8to5</i>	0.146*** (0.0233)	0.108*** (0.0185)	0.109*** (0.0189)	0.104*** (0.0172)
Spouse PT	0.0128 (0.0086)	0.0065 (0.0082)	0.0068 (0.0082)	0.0047 (0.0082)
Spouse FT	−0.0465*** (0.0088)	−0.0521*** (0.0088)	−0.0509*** (0.0083)	−0.0504*** (0.0083)
Spouse PT × <i>ratio8to5</i>	−0.0296** (0.0117)	−0.0356** (0.0110)	−0.0357** (0.0111)	−0.0263** (0.0110)
Spouse FT × <i>ratio8to5</i>	−0.0550*** (0.0113)	−0.0561*** (0.0113)	−0.0572*** (0.0110)	−0.0483*** (0.0110)
Observations	68,281	68,281	68,267	66,640

Notes: Standard errors are clustered at the occupation level and given in parentheses. ** $p < 0.05$, *** $p < 0.001$. Column (2) includes the average education level in the occupation as an additional control. Additional controls include a quartic in age, dummies for different levels of educational attainment, race dummies and year dummies. Column (3) also includes the share of male workers with at least some college education in the occupation who work more than 50 hours per week. Column (4) also includes occupation-level skill measures such as ‘social skills’, ‘abstract skills’, ‘routine skills’ and ‘manual skills’ constructed from the O*NET database.

argument is that there is a systematic wage penalty associated with doing household care related to our coordination measure. While we focus on the gender wage gap in our paper, the important message here is that the work-family conflict we identify is more widely applicable to all parents with care responsibilities.

In the Online Appendix, we conduct various robustness exercises in support of our main results. One concern with our measure is whether it is confounded with the prevalence of evening and night shifts. Nurses, for example, work shifts and are at work during all hours of the day, and the occupation would have low coordination requirements according to our measure. However, this may just be reflecting a 24-hour production cycle. To address this issue, in Table A.8 of the Online Appendix we add the fraction of workers who report being shift workers based on the 2004 CPS work schedule supplement as an additional control. Our results are robust to these additional controls. Table A.9 in the Online Appendix also reports regression results using our alternative measure of concentrated hours based on the Herfindahl index. These results are qualitatively very similar. An additional confirmation of our point is given in Cubas *et al.* (2021), in which we estimate the wage penalty for women using an alternative measure of the interruptions of household care activities during prime hours of the workday.²³

5. Quantitative Analysis

To assess the quantitative predictions of the model, we calibrate the model using aggregates from the US labour market. We restrict the analysis to 22 major SOC occupations (not including the military).²⁴ We also restrict the sample to married men and women

²³ While our analysis is static and relies on cross-sectional data, in Section A.3 of the Online Appendix we study earnings dynamics of females around childbirth. Specifically, we estimate the effect on earnings per hour of having a child and show that in occupations with a high *ratio8to5* the drop in earnings per hour after childbirth is more persistent.

²⁴ We use the number of full-time workers aged 18–65 married with children in the ATUS as weights to aggregate our detailed census 2002 occupation level measures to 2002 major SOC categories.

with children in the household. Among other variables of interest, solving the model yields *ratio8to5* for work and home care, as well as earnings for men and women in each occupation.

5.1. Calibration

We assume that the distribution of tastes is Fréchet with a common dispersion parameter.²⁵ Thus, we assume that, for an occupation j and a gender s , taste shocks are drawn from²⁶

$$F(\theta_{j,s}) = \text{Prob}(\theta_{j,s} \leq \theta_0) = \exp(-T_{j,s}\theta_0^{-\xi}).$$

The calibration chooses values for a total of 114 parameters:

$$(\{\alpha_j\}_{j=1}^{22}, \{\kappa_j\}_{j=1}^{22}, \{T_{j,m}\}_{j=1}^{22}, \{T_{j,f}\}_{j=1}^{22}, \{A_j\}_{j=1}^{22}, \rho, \nu^f, \nu^m, \beta).$$

We follow Hsieh *et al.* (2019) and set the value of β to 2/3, which implies an elasticity of substitution across occupations equal to 3. Because the dimension of the parameter vector is large, we set the vector $\{\kappa_1, \dots, \kappa_{22}\}$ equal to the vector of observed labour shares: the vector of the share of earnings of occupation j in total earnings.²⁷ Of course, with a CES production function, the model-implied labour shares differ from those in the data (because, with a CES, the value of κ_j is not equal to the labour share of occupation j). We provide a comparison of the model-implied and empirical labour shares below and show that the differences are not too large. The remaining parameter values are chosen to minimise the distance between the moments in the data and those generated by the model. The moments we pick to match are the following: the bunching ratios, *ratio8to5*, the fraction of females relative to males within an occupation, the fraction of employment in each occupation, the average earnings per hour for each occupation, the fraction of working time for males and females and the ratio of the average *ratio8to5* (work bunching ratio) to the average bunching ratio of home care.²⁸ The value of the last moment is largely influenced by ρ . A high value of ρ implies a low home care ratio (little home care takes place during prime time).²⁹

Table 8 shows the values for the occupation-specific moments we match. Table 9 illustrates the model fit by showing the correlation between the targeted moments in the data and in the model, as well as the mean absolute deviation in percentage terms between the moments in the data and in the model. The model fit is quite good. In addition, the table also shows the value of *ratio8to5* by gender, both in the data and the model. Empirically, women on average have a higher *ratio8to5* than men. The model delivers this result and the intuition is that, because women work fewer hours overall, they try to increase, relatively, the amount supplied during 8 to 5 to avoid as much as possible the productivity loss. This increases females' *ratio8to5* relative to that of males. The parameter values we obtain are shown in Table 10. The most interesting set of parameters are the α . Their distribution is rather skewed and their correlation with *ratio8to5*

²⁵ This assumption is typical in discrete choice models and made for tractability.

²⁶ Because the paper is not concerned with the distribution of tastes within occupations, setting a common dispersion parameter is irrelevant. We could assume either a different common dispersion parameter or a different dispersion parameter by occupation and gender. Doing so would yield different values for the (female) Fréchet parameters driving the mean for the model to be consistent with the empirical female shares across occupations.

²⁷ Calibrating the remaining 91 parameters is in itself a challenging computational problem.

²⁸ The bunching ratio for home care is defined similarly to *ratio8to5*, but counting the fraction of home care time between the prime working hours.

²⁹ Because we only model two activities and we normalise the length of each period to be 0.5, the model cannot deliver either work or home care bunching ratios in levels. Therefore we target the ratio.

Table 8. *Moments.*

Occupation no.	Occupation	Labour share	<i>ratio8to5</i>	Av. earn. per hour	% Fem.
1	Management	0.185	0.807	1.00	0.31
2	Business and financial operations	0.062	0.856	0.90	0.52
3	Computer and mathematical	0.053	0.837	1.08	0.22
4	Architecture and engineering	0.042	0.825	1.03	0.08
5	Life, physical, and social science	0.014	0.831	0.96	0.34
6	Community and social service occupations	0.016	0.778	0.67	0.54
7	Legal	0.021	0.863	1.09	0.46
8	Education, training and library	0.069	0.834	0.72	0.72
9	Arts, design, entertainment, sports and media	0.014	0.817	0.82	0.33
10	Healthcare practitioners and technical	0.068	0.723	0.88	0.70
11	Healthcare support	0.009	0.710	0.42	0.87
12	Protective service	0.030	0.592	0.73	0.12
13	Food preparation and serving related	0.012	0.604	0.37	0.46
14	Building and grounds cleaning and maintenance	0.017	0.715	0.40	0.31
15	Personal care and service	0.008	0.667	0.42	0.73
16	Sales and related	0.091	0.788	0.72	0.34
17	Office and administrative support	0.085	0.826	0.54	0.72
18	Farming, fishing and forestry	0.004	0.627	0.33	0.24
19	Construction and extraction	0.055	0.791	0.62	0.01
20	Installation, maintenance and repair	0.042	0.764	0.65	0.03
21	Production	0.057	0.648	0.52	0.23
22	Transportation and material moving	0.045	0.659	0.51	0.11

Notes: The table presents the occupational level moments we use in our calibration. Labour shares are calculated by dividing the total earnings of workers in each occupation by the total mass of earnings in the sample. The *ratio8to5* is our measure of coordination using time use data obtained as we explain in the text. We also report the average earnings per hour of workers in each of the occupations (Av. earn per hour) and the share of females in the total number of workers in each occupation (% Fem.).

is -0.5 . In other words, the *ratio8to5* across occupations is mainly determined by the α , but not completely (otherwise, the two would be perfectly negatively correlated). The share of females in an occupation also plays an important role. The *ratio8to5* of females is higher than that of males, so if an occupation is 90% female (as is, for example, healthcare support) it must have a larger α than an occupation with the same *ratio8to5* but, say, only 30% females. The skewness is an artefact of coordination costs being virtually zero for a large α .³⁰ It is also worth noting that we estimate a relatively low value of ρ (0.48), which implies that time allocations to household care in different periods are less than perfect substitutes for each other.

As a way to validate the model, in Table 11 we re-run the regressions reported in Section 4, but we now estimate them for 22 occupations using individual data and model-generated data. The first column displays the coefficients on the female dummy, *ratio8to5*, and the interaction between the two using data.³¹ The second column shows the analogous coefficients from our model-simulated data. Note that the regression coefficients were not a targeted moment in the calibration. The *ratio8to5* coefficient is 0.29 in the model, and it is larger than that in the data because there are fewer elements affecting earnings per hour in the model.³² The coefficient on

³⁰ There is little information about coordination costs for an α that exceeds 50, as in the case of 'Protective services'; it basically has the same effect of an α that is larger than 5. This is a special occupation and it could be that our moment-matching function is not correctly identifying the value of α for it.

³¹ Note that the coefficients will not exactly match those in Table 6 due to the fact that our occupation measure is aggregated to 22 groups.

³² The values of the coefficient differ because of the heterogeneity over many dimensions (age, education, etc.) across individuals within occupations, whereas in the model the heterogeneity is only between males and females.

Table 9. *Model Fit.*

<i>Panel A: occupational-level moments</i>		
Moment		Corr. coefficient, model and data
<i>ratio8to5</i>		1.00
Average earnings per hour		1.00
% Females		0.98
Occupational shares		0.98
<i>Panel B: economy-wide moments</i>		
Moment	Data	Model
Ratio hours worked male-female	1.20	1.24
<i>ratio8to5</i> work/ <i>ratio8to5</i> household care.	2.03	2.22
<i>Panel C: overall measure of fit</i>		
Average absolute % deviation (model and data)	5.30%	
<i>Panel D: non-targeted moments</i>		
Moment	Data	Model
<i>ratio8to5</i> males	0.74	0.62
<i>ratio8to5</i> females	0.82	0.70

Notes: The table shows the model fit by comparing the value of the targeted moments in the data and in the model. In addition, panel D shows the values of non-targeted moments. For the occupational-level moments, we show their values in the data and in the model (panel A). For the economy-wide targeted moments, we show in panel B, for each targeted moment, the correlation across occupations between the value of the moments in the data and in the model. The last line of the table gives an overall measure of fit (average absolute percentage deviation between the model and data).

the interaction between *ratio8to5* and the female dummy is -0.04 . What drives the positive relationship between the gender gap and *ratio8to5* in the model? The coordination cost is higher the lower α , which translates into a higher *ratio8to5*. Since females supply more home care, and home care is not perfectly substitutable across hours of the day, they supply fewer market hours during the prime period. As a result, their effective hours drop and their compensation reflects the lost hours.

As a way to further validate the model, we examine its performance on other non-targeted moments. For example, the model implies that, for high- α occupations (lower coordination costs), the ratio of males' to females' labour supply is lower. The reason is that females supply more labour in low- α occupations because it is costly not doing so. The data seem to be in line with this model's prediction. The correlation of the ratio of males' to females' labour supply across occupations between data and the model is 0.26. In the model, females' *ratio8to5* are higher than males' *ratio8to5*. The reason is that it is costly to supply labour outside of prime time, so females proportionally supply more labour during that period. This increases their *ratio8to5*. The relative ratio of males' *ratio8to5* to females' work bunching ratio is 0.89 in the model, while it is 0.90 in the data.

5.2. *The Baseline Economy*

Solving the model for the set of calibrated parameter values delivers an equilibrium that features an allocation of males and females over occupations and a vector of occupation-specific wage rates compensating a unit of labour. The equilibrium also generates a gender wage gap in

Table 10. *Parameter Values.*

Panel A: occupational-specific parameters						
Occupation no.	Occupation	κ	α	A	T_f	T_m
1	Management	0.185	0.98	0.89	8.05	1.78
2	Business and financial operations	0.062	0.75	1.81	5.86	0.53
3	Computer and mathematical	0.053	0.65	2.12	1.51	0.65
4	Architecture and engineering	0.042	0.64	2.26	0.49	0.65
5	Life, physical and social science	0.014	0.77	3.87	0.77	0.18
6	Community and social service occupations	0.016	1.46	2.91	2.45	0.25
7	Legal	0.021	0.63	3.41	1.19	0.17
8	Education, training and library	0.069	1.24	1.51	15.55	0.14
9	Arts, design, entertainment, sports and media	0.014	0.88	3.58	0.87	0.21
10	Healthcare practitioners and technical	0.068	2.50	1.41	11.31	0.04
11	Healthcare support	0.009	2.72	2.60	4.21	0.02
12	Protective service	0.030	68.64	1.70	0.65	0.68
13	Food preparation and serving related	0.012	6.92	1.96	3.12	0.52
14	Building and grounds cleaning and maintenance	0.017	2.00	2.01	2.94	0.81
15	Personal care and service	0.008	3.31	2.59	2.80	0.11
16	Sales and related	0.091	1.21	1.05	7.33	1.35
17	Office and administrative support	0.085	1.37	1.28	34.97	0.00
18	Farming, fishing and forestry	0.004	3.90	3.17	0.53	0.26
19	Construction and extraction	0.055	0.82	1.41	0.17	1.88
20	Installation, maintenance and repair	0.042	1.07	1.59	0.28	1.33
21	Production	0.057	3.20	1.05	4.23	1.89
22	Transportation and material moving	0.045	2.69	1.13	1.52	1.71
Panel B: rest of the parameters						
ρ		0.48				
ν_f		0.43				
ν_m		0.57				

Notes: Panel A shows the values of the parameters that are specific to the different occupations and panel B shows the values obtained for the utility function, ν_m and ν_f , for males and females, respectively. In addition, panel B presents the value obtained for the parameter that governs the elasticity of substitution of the technology for household care, ρ .

Table 11. *Regressions: Model versus Data.*

	Data	Model
Female	−0.270*** (0.003)	−0.06
ratio8to5	0.097*** (0.002)	0.29
Female × ratio8to5	−0.018*** (0.004)	−0.04

Notes: Standard errors are given in parentheses. *** $p < 0.01$. Note that this table shows the estimates of the regression using individual data for married workers with children (the data column) for 22 occupations and the estimates of the same regression using data generated by the model in its baseline calibration (the model column). The dependent variable is earnings per hour.

each occupation and an economy-wide gender wage gap. The economy-wide gender wage gap can be decomposed into a between- and a within-occupation component. More specifically, let the earnings ratio between males and females for the whole economy be defined as $egap = e_m/e_f$, where e_m and e_f represent the mean earnings of males and females across occupations,

Table 12. *Gender Earnings Gap (%)*.

	Overall	Between	Between (sorting)	Within
Data	23.2	−0.9	−	24.1
Baseline	8.9	1.7	−	7.2
Equal α ($\alpha = 2.72$)	6.4	4.4	0.2	2.0
50% drop in $\nu_m - \nu_f$	6.6	3.0	0.4	3.6
Increase in ρ	9.7	4.7	1.2	5.0

respectively. That is,

$$egap = \frac{e_m}{e_f} = \frac{\sum_{j=1}^J \gamma_{m,j} e_{m,j}}{\sum_{j=1}^J \gamma_{f,j} e_{f,j}},$$

where $\gamma_{m,j}$ and $\gamma_{f,j}$ are the proportions of males and females in occupation j over total males and females in the population. Thus,

$$egap = \frac{\sum_{j=1}^J \gamma_{m,j} e_{f,j}}{\sum_{j=1}^J \gamma_{f,j} e_{f,j}} \frac{\sum_{j=1}^J \gamma_{m,j} e_{m,j}}{\sum_{j=1}^J \gamma_{m,j} e_{f,j}}.$$

The first term of the right-hand side of the equation represents the *between* component, whereas the second represents the *within* component. In the tables we report each component as a log ratio.³³

Table 12 reports the results. The first row contains the decomposition in the data. The overall gender wage gap is 23.2%. The within component is 24.1% while the between component is −0.9%. This is broadly consistent with Goldin (2014), who found that the bulk of the gender wage gap exists within occupations and only a small component is due to the between portion.

In the baseline model (second row) the gender wage gap is 8.9%; this moment was not targeted. The model predicts a within component of 7.2% that accounts for 30% of the within component in the data. The model also generates a between component of 1.7%. The endogenous channel in our model—the interaction between preferences and coordination costs—has implications for the both the within and between components. Although this channel influences mostly the within component, it also affects the between component because it affects how women sort into occupations. While it explains a substantial component of the within component, it does not account for the majority, indicating that there are other forces in the economy that affect the gender wage gap within an occupation.

5.3. *Counterfactual Experiments*

In this section we conduct counterfactual experiments to assess the impact of various parameters on the gender wage gap. The key parameters of interest are the α that reflect coordination costs, the ν that reflect preferences for consumption and for household care, and ρ that determines the elasticity of substitution between household care at different times of the day.

³³ There is not a unique way to perform this decomposition. For example, you could fix the share of males in the occupations instead of females. The magnitudes can slightly change, but the results are qualitatively the same.

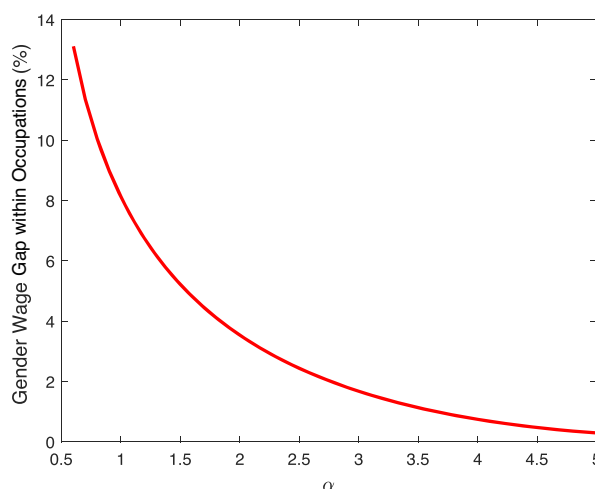


Fig. 5. *Gender Earnings Gap within Occupations: The Effect of α .*

Notes: The figure shows the value of the within component of the gender wage gap (y axis) when the parameter α (x axis) is equal for every occupation and takes values from 0.6 (the minimum estimated value for our baseline economy) to 5.

5.3.1. *Coordination of schedules and the gender wage gap*

In the first experiment we set α to be equal across occupations. We set α at a relatively high value of 2.72 (reflecting low coordination costs), which is the value estimated for ‘Health care support’. One motivation for such an experiment is the introduction of a new technology such as on-line connections and internet technology. The effect is lower costs of coordinating with other workers. Since in this experiment women still have a higher preference for household care (lower ν), everything else equal, they work less and allocate more hours to home production relative to men. However, the costs of doing so are lower.

The gender gap falls from 8.9% (baseline) to 6.3%. As shown in the third row of Table 12, the within component falls from 7.2% to 2.0%. The within component falls substantially because, with a relatively high α , the penalty for not working during prime time is lower. As a result, despite women’s larger supply of household care, their earnings per hour are now much closer to those of males. Figure 5 shows the within component of the gender gap (vertical axis) when this counterfactual is repeated for different values of α (horizontal axis). Low values of α , such as that for ‘Management’ for example, imply a large within component of around 10%. As α becomes larger, the coordination costs become negligible and the within-occupation gender gap approaches zero. Figure 5 shows that there is little difference between moderately high α ’s and very high α ’s; that is, once α reaches a value of 5 and greater, the within-occupation gender gap essentially disappears.

The between component rises when we equalise the α across occupations. Both males and females move to occupations with initially low α ’s because the coordination penalty is now lower. These occupations are even more attractive to women, so the female share rises in these occupations. This effect by itself increases the between component, because low- α occupations tend to pay more. To assess the size of this effect, in Table 12 we separately report the gender wage gap between occupations holding earnings and occupation sizes fixed at their baseline values.

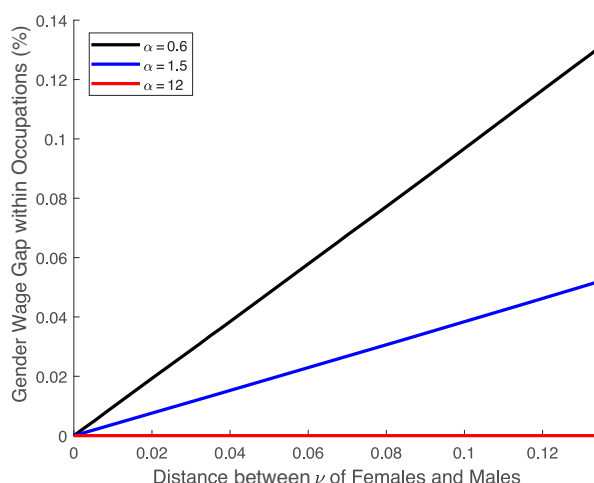


Fig. 6. *Gender Earnings Gap within Occupations: The Effect of ν .*

Notes: The figure shows the value of the within component of the gender wage gap (y axis) when we change the difference between females and males in the value of parameter ν (x axis) for different values of α . A value of 0 on the x axis indicates that $\nu_m = \nu_f$.

This column, which we label ‘sorting’, isolates the effect of rising female shares in initially low- α occupations. In this case the between-occupation gender gap is 0.2. However, in equilibrium, earnings and occupation sizes also change. Low- α occupations get larger and their earnings rise (because the coordination costs are now smaller in these occupations). Because the fraction of men is larger in high-earning occupations, and these occupations increase in size and earnings because now α is higher, the between component of the gender gap rises.

5.3.2. A change in female household care hours

In this experiment we reduce the difference between male and female preferences for household care. One possible interpretation is that a change in social norms equalises the household care responsibilities of males and females. We reduce the gap in the ν by 50% by reducing ν_m to 0.535 and raising ν_f to 0.465. The within component falls from 7.2% to 3.6%. The reason for the fall is that an increase in ν_f increases the amount of work during prime time. This lowers the penalty that females face.

An alternative way of looking at this counterfactual is shown in Figure 6. The horizontal axis measures the distance between the ν (a value of 0.14 is equal to the baseline and a value of 0 means that $\nu_m = \nu_f$). On the vertical axis we measure the within component of the gender gap. As the distance between the ν drops, the within component goes to zero. The rate at which it drops to zero depends on the occupation. As occupations are defined by their α , we plot the within component against the within component for occupations with a small value of α , 0.6; a middle value, 1.5 and a high value, 12. When α is large, i.e., coordination costs are low, the within component is virtually zero even when women supply substantially more home care than men. For an occupation such as ‘Architecture and engineers’ with ($\alpha = 0.64$), then the within-gender gap is low only when preferences between males and females are similar.

Interestingly, the between component rises. As a result of the ν changing, women are now more likely to move into low- α occupations. This sorting effect alone reduces the between-occupation gender gap from 1.7 in the baseline case to 0.4. However, in equilibrium, earnings and occupation sizes change. In this case, because women now prefer to work more, their labour supply rises. This happens in all occupations, but the effect is bigger in occupations that are relatively more populated by women. As a result, wage rates in female-intensive occupations fall (responding to the larger supply), leading to an increase in the between component from 1.7 to 3.0.

5.3.3. *A change in the ability to smooth household care during the day*

In our model, the value of ρ determines the elasticity of substitution between household care time during the two parts of the day. A high value of ρ means that it is relatively easy to substitute household care activities throughout the day. In our baseline calibration $\rho = 0.48$, which indicates that household care activities are fairly substitutable, but imperfectly so. Although the timing of these activities may be difficult to change (reflecting an imperfect degree of substitution), someone other than the parent could be responsible for undertaking them. For example, curricular education normally takes place in a school during normal business hours, but parents outsource that activity to school teachers and staff. As shown in Section 1.2, we focus on care (i.e., doctor's appointments, school meetings, homework supervision) that may be hard to outsource. What exactly determines the degree of substitution is not clear. One interpretation is that there are constraints on the time of the day in which some activities take place. For example, an appointment with a school teacher normally takes place before 5 p.m. An alternative interpretation is that they reflect parenting styles of modern societies, a recent phenomenon that has been extensively studied, for instance, by Doepke and Zilibotti (2019).

In order to study the effect of changes in the ability to substitute household care time during the day, we perform a counterfactual exercise in which we increase ρ to 0.65. As in our baseline case, women put more value on household care activities and allocate more time to household care relative to men. The main difference is that, compared to the baseline case, women can now more easily distribute the household care to off-peak times so they do not incur a productivity loss. As a result, the within-gender wage gap decreases from 7.2% to 5.0%, as predicted. The increase in ρ has little effect on sorting. However, there is again a substantial increase in the between-occupation gender gap in equilibrium. Productivity and earnings rise in low- α occupations. Since men initially had higher representation in these occupations, the between-occupation component of the gender wage gap rises.

6. Final Remarks

Although women have made remarkable gains in the labour market over the past five decades, there is still a substantial gap in their earnings relative to men. Most of the unexplained gap is associated with earnings gaps that arise within occupations. In this paper we explore a mechanism that can explain why the gender gap differs across occupations.

Central to our analysis is the joint decision of workers to allocate time to market work and to household care. Using time diary data, we document that married women with children who report being full-time workers, work less on the job and do more household care than their male counterparts. We also document that occupations vary in the degree to which total hours worked in the occupation are concentrated during peak hours of the day—a measure that we interpret as reflecting the degree of coordinated work schedules in the occupation. Our measure

of a (in)flexible work schedule is therefore distinct from other papers in the literature that focus on the quantity of hours worked. We find that, while men and single women receive a wage premium in occupations with concentrated schedules, married women with children much less of one. Conditional on being in an occupation, less working time (more household care time) at peak hours of the day entails a productivity loss, and thus earnings are lowered for women relative to men. We calibrate our model to US data and show that the greater demand for household care time by women, together with the coordination of work time required in different occupations generates a gender wage gap of 8.9%, which corresponds to approximately 40% of the observed gender earnings gap among married men and women with children. As in the data, most of the gender wage gap is within occupations. The value predicted by the model is 7.2%, which accounts for 30% of that observed in the data. If occupation-level coordination was set equal to the level of ‘Health care support’—an occupation with relatively low coordination, the within-occupation gender gap due to women’s higher demand for household time falls by more than half to 2.0%.

Our paper provides new insights by studying the interplay of family constraints and the coordination of schedules and its effect on women’s wages relative to men’s. We present a rich static model, but the constraints we study in this paper may well have dynamic implications. For example, the inability to provide hours during peak times may result in less human capital accumulation for women, such that their wage disadvantage may persist even well after childrearing, amplifying the effect of the coordination friction on the gender wage gap. We leave these important topics for future research.

University of Houston, USA

University of Houston, USA

Temple University, USA

Additional Supporting Information may be found in the online version of this article:

Online Appendix Replication Package

References

- Acemoglu, D. and Autor, D. (2011). ‘Skills, tasks and technologies: Implications for employment and earnings’, in (D. Card and O. Ashenfelter, eds), *Handbook of Labor Economics*, vol. 4, pp. 1043–171, Amsterdam: Elsevier.
- Albanesi, S. and Olivetti, C. (2009). ‘Production, market production and the gender wage gap: Incentives and expectations’, *Review of Economic Dynamics*, vol. 12(1), pp. 80–107.
- Altonji, J.G. and Paxson, C.H. (1988). ‘Labor supply preferences, hours constraints, and hours-wage trade-offs’, *Journal of Labor Economics*, vol. 6(2), pp. 254–76.
- Altonji, J.G. and Paxson, C.H. (1992). ‘Labor supply, hours constraints, and job mobility’, *Journal of Human Resources*, vol. 27(2), pp. 256–78.
- Battiston, D., Blanes i Vidal, J. and Kirchmaier, T. (2020). ‘Face-to-face communication in organizations’, *The Review of Economic Studies*, vol. 88(2), pp. 574–609.
- Bick, A. and Fuchs-Schündeln, N. (2018). ‘Taxation and labour supply of married couples across countries: A macroeconomic analysis’, *Review of Economic Studies*, vol. 85(3), pp. 1543–76.
- Blundell, R., Brewer, M. and Francesconi, M. (2008). ‘Job changes and hours changes: Understanding the path of labor supply adjustment’, *Journal of Labor Economics*, vol. 26(3), pp. 421–53.
- Bureau of Labor Statistics. (2014). ‘American time use survey’, <https://www.bls.gov/tus/data.htm>.
- Cardoso, A.R., Hamermesh, D.S. and Varejao, J. (2012). ‘The timing of labor demand’, *Annals of Economics and Statistics*, vol. 105/106, pp. 15–34.
- Cha, Y. and Weeden, K.A. (2014). ‘Overwork and the slow convergence in the gender gap in wages’, *American Sociological Review*, vol. 79(3), pp. 457–84.

- Chen, M.K., Chevalier, J.A., Rossi, P.E. and Oehlsen, E. (2019). 'The value of flexible work: Evidence from uber drivers', *Journal of Political Economy*, vol. 127, pp. 2735–94.
- Chetty, R., Friedman, J.N., Olsen, T. and Pistaferri, L. (2011). 'Adjustment costs, firm responses, and micro vs. macro labor supply elasticities: Evidence from Danish tax records', *The Quarterly Journal of Economics*, vol. 126(2), pp. 749–804.
- Cortes, P. and Pan, J. (2016a). 'Prevalence of long hours and skilled women's occupational choices', Boston University, Working paper.
- Cortes, P. and Pan, J. (2016b). 'When time binds: Returns to working long hours and the gender wage gap among the highly skilled', Boston University, Working paper.
- Cubas, G., Juhn, C. and Silos, P. (2021). 'Work-care balance over the day and the gender wage gap', *AEA Papers and Proceedings*, vol. 111, pp. 149–53.
- Deming, D.J. (2017). 'The growing importance of social skills in the labor market', *The Quarterly Journal of Economics*, vol. 132(4), pp. 1593–640.
- Denning, J.T., Jacob, B.A., Lefgren, L. and vom Lehn, C. (2019). 'The return to hours worked within and across occupations: Implications for the gender wage gap'. NBER Working Paper, No. 25739, April 2019.
- Doepke, M. and Tertilt, M. (2016). 'Families in macroeconomics', in (J.B. Taylor and H. Uhlig, eds.), *Handbook of Macroeconomics*, vol. 2, pp. 1789–891, Amsterdam: Elsevier.
- Doepke, M. and Zilibotti, F. (2019). *Love, Money, and Parenting: How Economics Explains the Way We Raise Our Kids*, Princeton, NJ: Princeton University Press.
- Duchini, E. and Effenterre, C.V. (2017). 'Do women want to work more or more regularly? Evidence from a natural experiment'. Working paper, Paris School of Economics.
- Eden, M. (2017). 'The week', Brandeis University, Working paper.
- Erosa, A., Fuster, L., Kambourov, G. and Rogerson, R. (2017). 'Hours, occupations, and gender differences in labor market outcomes', Working Paper 23636, National Bureau of Economic Research.
- Flabbi, L. and Moro, A. (2012). 'The effect of job flexibility on female labor market outcomes: Estimates from a search and bargaining model', *Journal of Econometrics*, vol. 168(1), pp. 81–95.
- Flood, S.M., Sayer, L.C. and Backman, D. (2022). 'American Time Use Survey data extract builder: Version 3.1 [dataset]', College Park, MD: University of Maryland and Minneapolis, MN: IPUMS, <https://doi.org/10.18128/D060.V3.1>.
- Gibbs, M., Mengel, F. and Siemroth, C. (2021). 'Work from home & productivity: Evidence from personnel & analytics data on it professionals', Discussion Paper 14336, Institute of Labor Economics.
- Gicheva, D. (2013). 'Working long hours and early career outcomes in the high-end labor market', *Journal of Labor Economics*, vol. 31(4), pp. 785–824.
- Goldin, C. (2014). 'A grand gender convergence: Its last chapter', *American Economic Review*, vol. 104(4), pp. 1091–119.
- Goldin, C. and Katz, L.F. (2011). 'The cost of workplace flexibility for high-powered professionals', *The Annals of the American Academy of Political and Social Science*, vol. 638(1), pp. 45–67.
- Guner, N., Kaya, E. and Sánchez-Marcos, V. (2014). 'Gender gaps in Spain: Policies and outcomes over the last three decades', *SERIEs: Journal of the Spanish Economic Association*, vol. 5(1), pp. 61–103.
- Guner, N., Kaya, E. and Sánchez Marcos, V. (2019). 'Labor market frictions and lowest low fertility', Discussion Paper 12771, Institute of Labor Economics.
- Hamermesh, D.S. (1999). 'The timing of work over time', *ECONOMIC JOURNAL*, vol. 109, pp. 37–66.
- Hsieh, C., Hurst, E., Jones, C.I. and Klenow, P.J. (2019). 'The allocation of talent and U.S. economic growth', *Econometrica*, vol. 87(5), pp. 1439–74.
- Labanca, C. and Pozzoli, D. (2022). 'Constraints on hours within the firm', *Journal of Labor Economics*, vol. 40, pp. 473–503.
- Mas, A. and Pallais, A. (2017). 'Valuing alternative work arrangements', *American Economic Review*, vol. 107(12), pp. 3722–59.
- O*NET, OnLine. (2020). 'National center for O*NET development', www.onetonline.org/help/online/data.
- Prescott, E.C., Rogerson, R. and Wallenius, J. (2009). 'Lifetime aggregate labor supply with endogenous workweek length', *Review of Economic Dynamics*, vol. 12(1), pp. 23–36.
- Rogerson, R. (2011). 'Individual and aggregate labor supply with coordinated working times', *Journal of Money, Credit and Banking*, vol. 43, pp. 7–37.
- Rogerson, R. and Wallenius, J. (2009). 'Micro and macro elasticities in a life cycle model with taxes', *Journal of Economic Theory*, vol. 144(6), pp. 2277–92.
- Rosen, H.S. (1976). 'Taxes in a labor supply model with joint wage-hours determination', *Econometrica*, vol. 44(3), pp. 485–507.
- Ruggles, S., Flood, S., Goeken, R., Schouweiler, M. and Sobek, M. (2022). 'IPUMS USA: version 12.0 [dataset]', Minneapolis, MN: IPUMS, <https://doi.org/10.18128/D010.V12.0>.
- Stewart, J. (2010). 'The timing of maternal work and time with children', *Industrial and Labor Relations Review*, vol. 64(1), pp. 181–200.
- Wasserman, M. (2019). 'Hours constraints, occupational choice, and gender: Evidence from medical residents'. (March 19, 2019). Available at SSRN: <https://ssrn.com/abstract=3371100>.
- Weiss, Y. (1996). 'Synchronization of work schedules', *International Economic Review*, vol. 37(1), pp. 157–79.
- Wiswall, M. and Zafar, B. (2017). 'Preference for the workplace, investment in human capital, and gender', *The Quarterly Journal of Economics*, vol. 133(1), pp. 457–507.